US 20250279346A1

(19) United States

(12) Patent Application Publication

(10) Pub. No.: US 2025/0279346 A1

(43) Pub. Date: Sep. 4, 2025

Kim et al.

(54) HIGH ASPECT RATIO VIAS WITH LOW STRESS IN INTEGRATED CIRCUIT PACKAGES

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **Kihyun Kim**, Chandler, AZ (US); **Micah David Armstrong**, Gilbert, AZ (US); **Darko Grujicic**, Chandler, AZ (US); **Shayan Kaviani**, Phoenix, AZ (US); **Rahul Nagaraj Manepalli**, Chandler, AZ (US); **Srinivas Venkata Ramanuja Pietambaram**, Chandler, AZ (US); **Dilan Seneviratne**, Phoenix, AZ (US); **Rengarajan Shanmugam**, Tempe, AZ (US); **Joshua James Stacey**, Chandler, AZ (US); **Marcel Arlan Wall**, Phoenix, AZ (US); **Ehsan Zamani**, Phoenix, AZ (US)

(21) Appl. No.: **18/593,337**

(22) Filed: **Mar. 1, 2024**

Publication Classification

(51) **Int. Cl.**

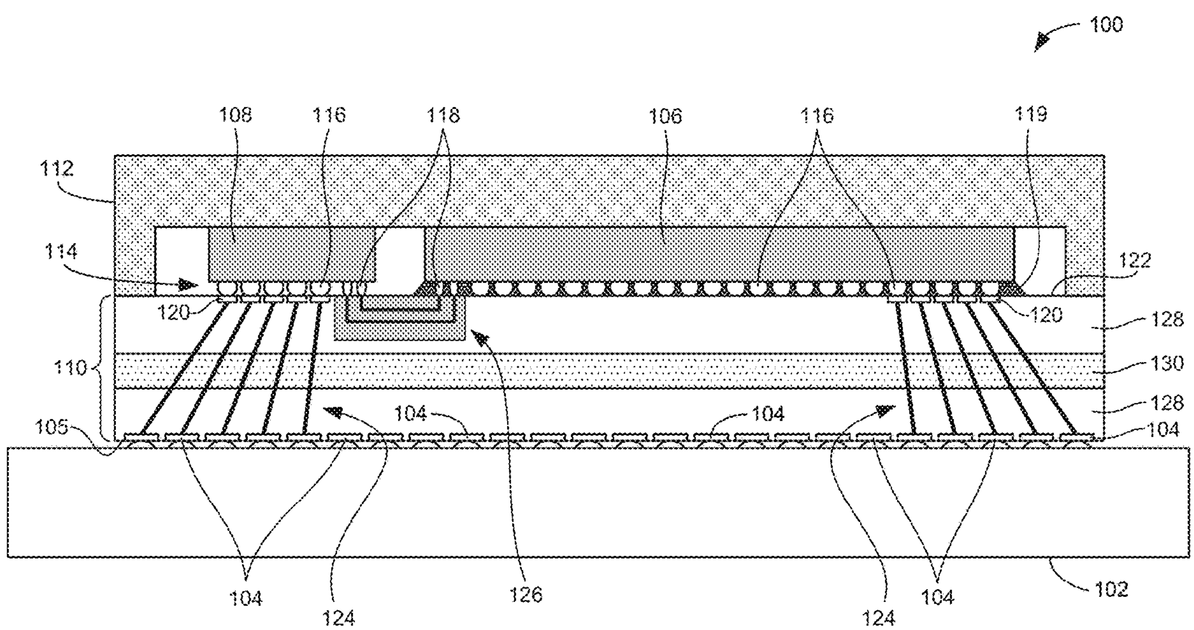
H01L 23/498	(2006.01)
H01L 23/00	(2006.01)
H01L 25/065	(2023.01)

(52) **U.S. Cl.**

CPC .. ***H01L 23/49838*** (2013.01); ***H01L 23/49822*** (2013.01); ***H01L 24/08*** (2013.01); ***H01L 24/16*** (2013.01); ***H01L 25/0655*** (2013.01); *H01L 2224/08146* (2013.01); *H01L 2224/08155* (2013.01); *H01L 2224/16146* (2013.01); *H01L 2224/16157* (2013.01); *H01L 2924/01029* (2013.01); *H01L 2924/1431* (2013.01); *H01L 2924/1434* (2013.01); *H01L 2924/351* (2013.01)

(57) **ABSTRACT**

High aspect ratio vias with low stress in integrated circuit packages are disclosed. A package substrate includes: a core having a via extending therethrough; a first conductive material in the via; a dielectric material at least partially between a wall of the via and the first conductive material; and a second conductive material in the via. The second conductive material is closer to a central region of the via than the first conductive material is to the central region of the via. The first conductive material is different from the second conductive material. The first conductive material is at least partially between the dielectric material and the second conductive material.



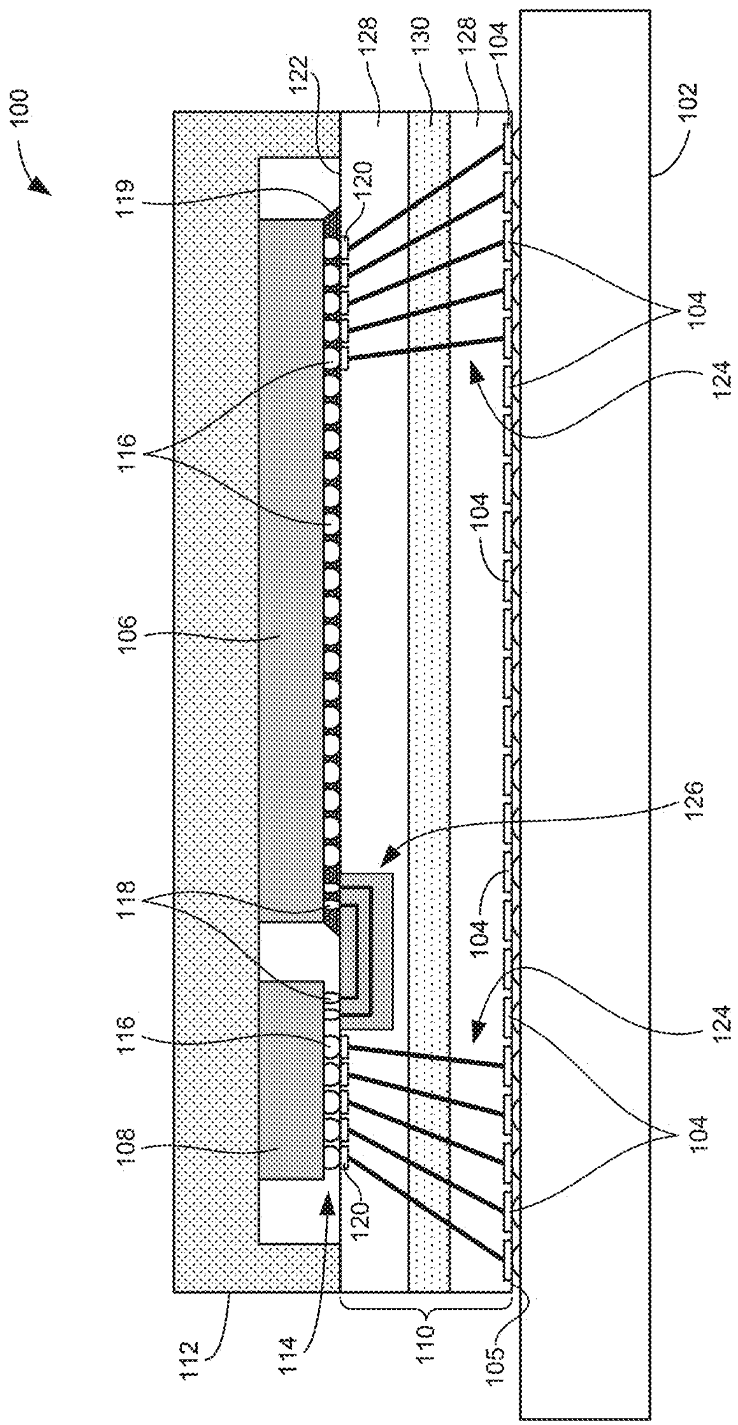

100
102
104
105
106
108
110
112
114
116
118
119
120
122
124
126
128
130

FIG. 1

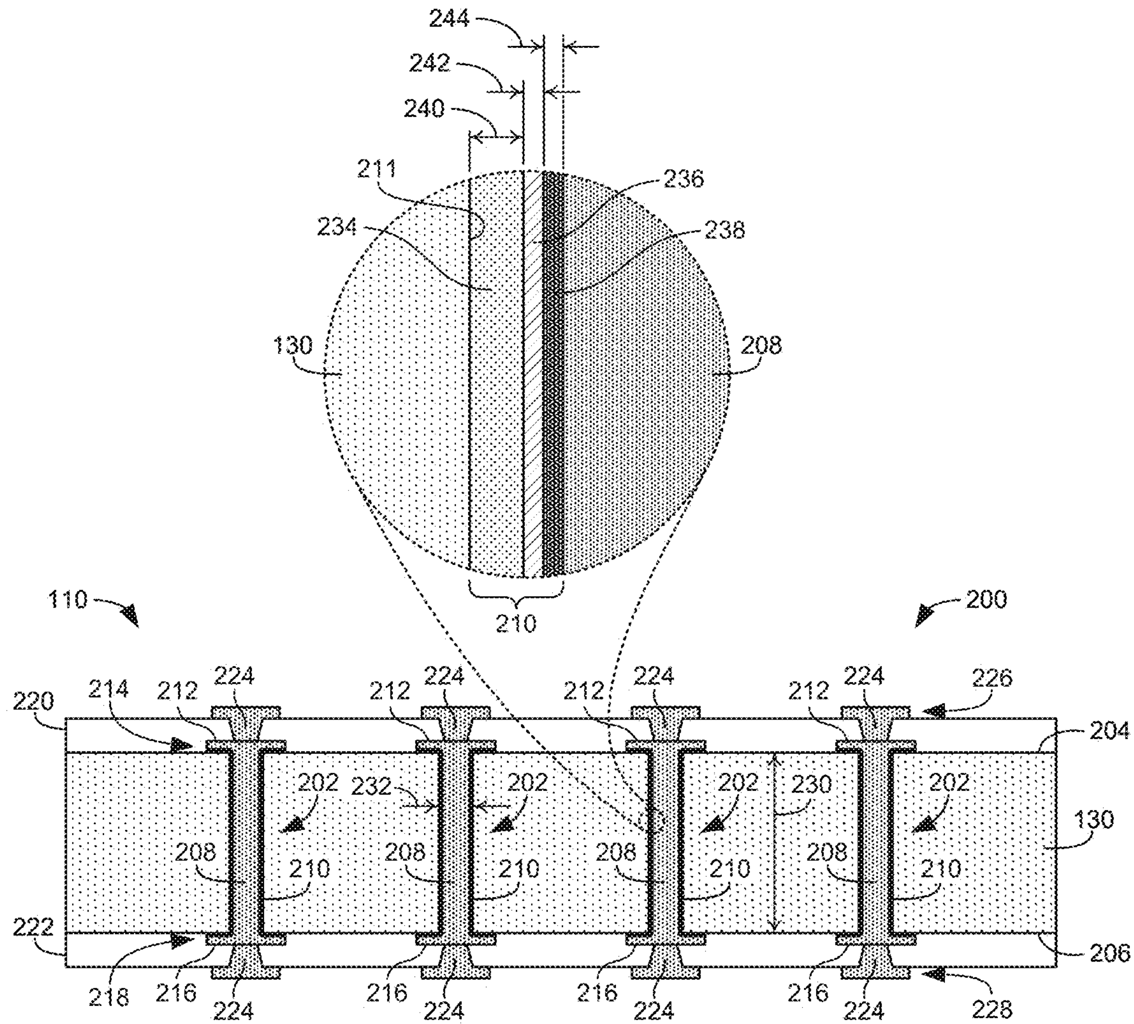

244
242
240
211
236
234
238
130
208
210
110
200
220
214
212
224
226
204
202
232
230
130
208
210
222
206
218
216
228

FIG. 2

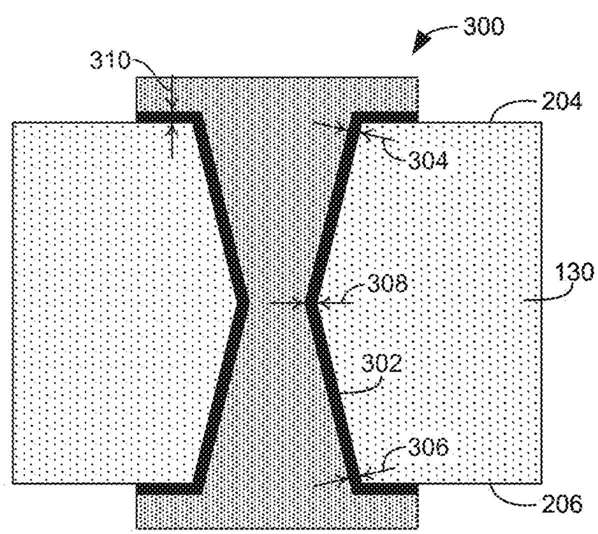

300
310
204
304
130
308
302
306
206

FIG. 3

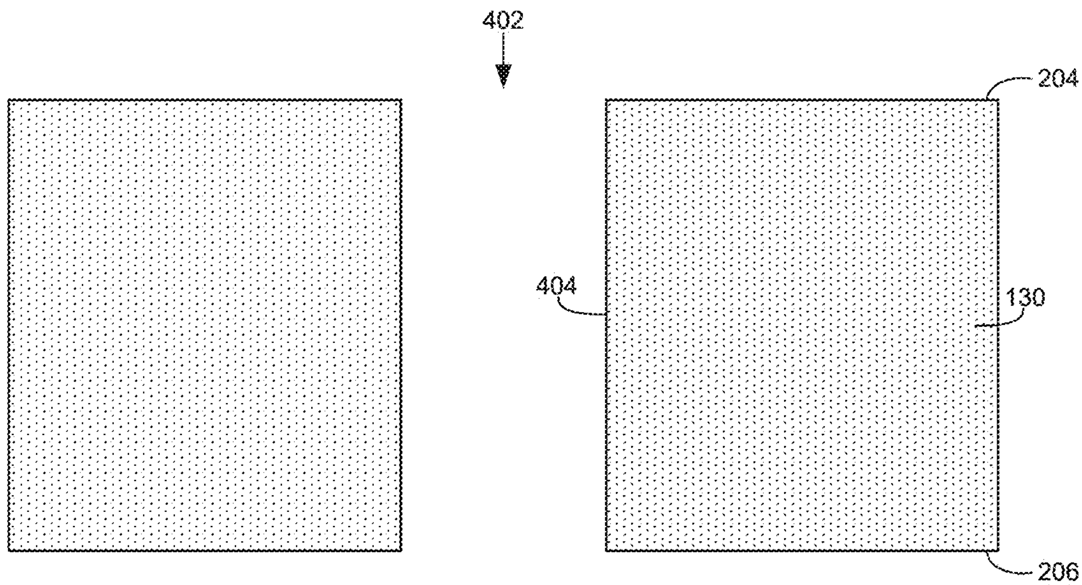


FIG. 4

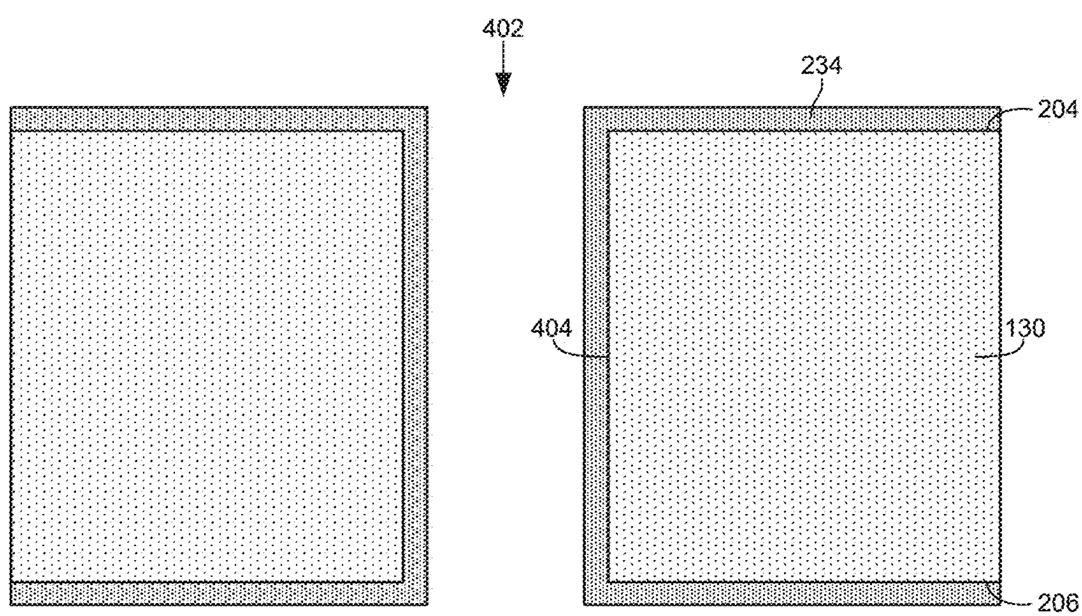


FIG. 5

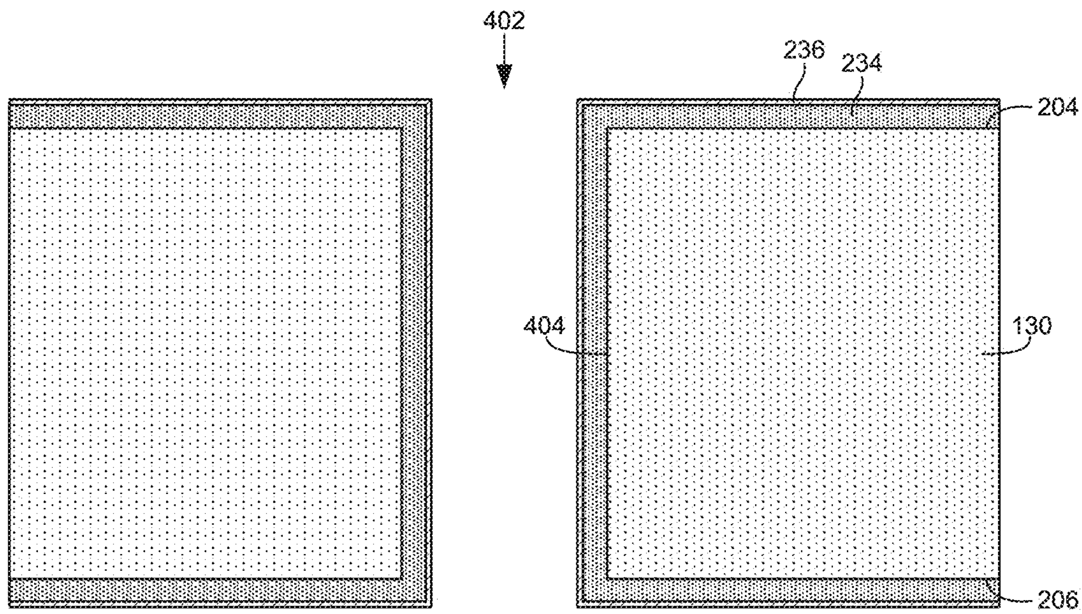


FIG. 6

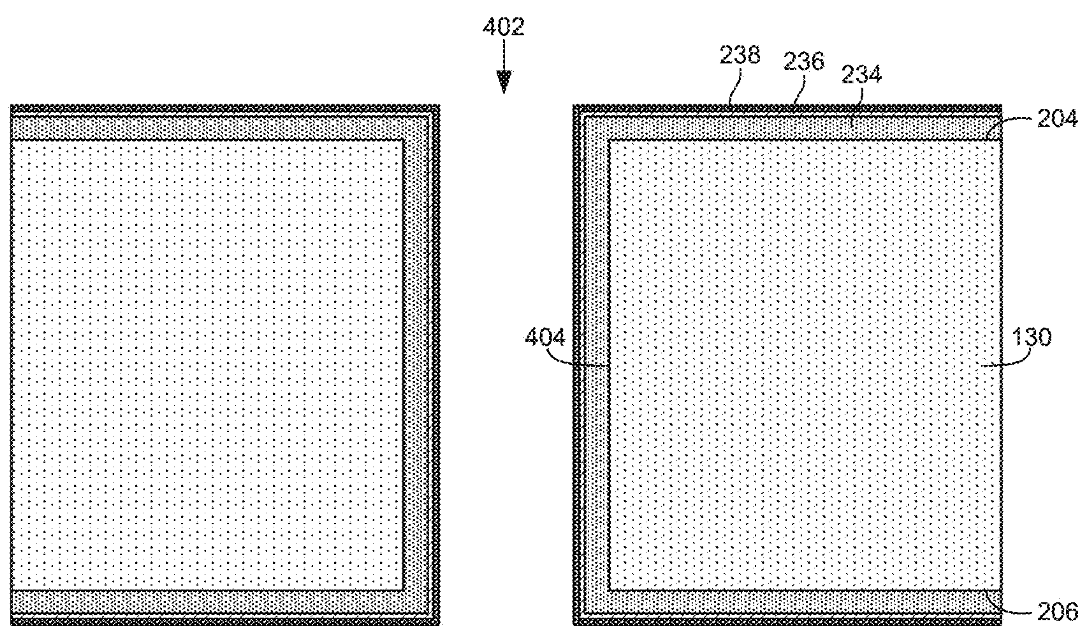


FIG. 7

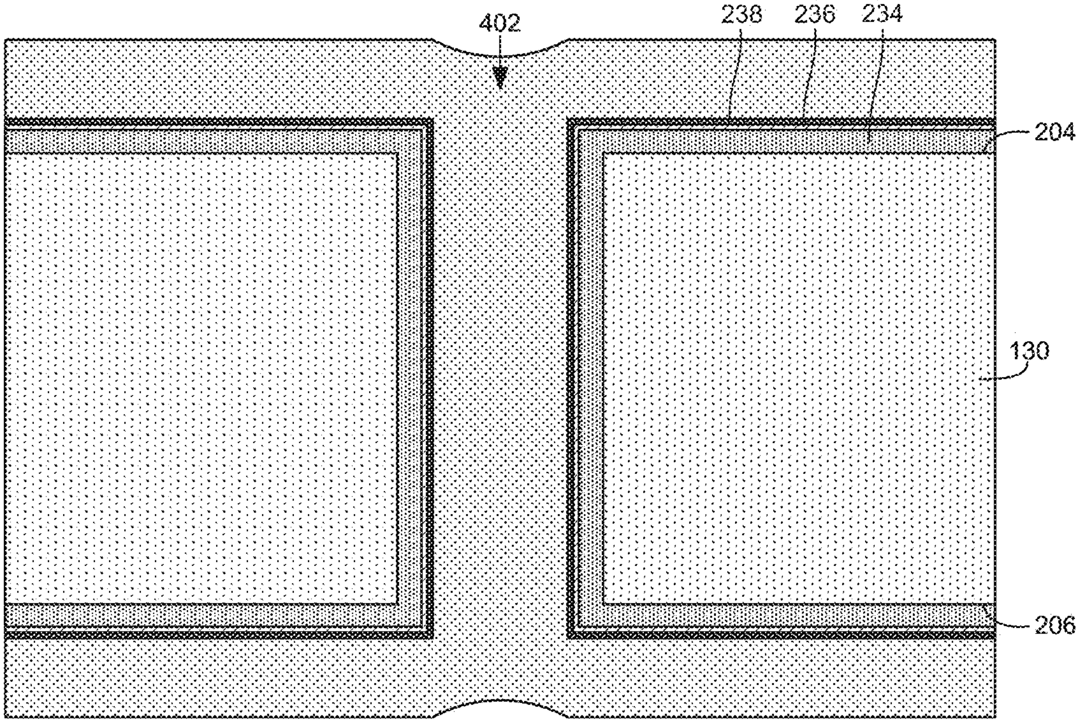


FIG. 8

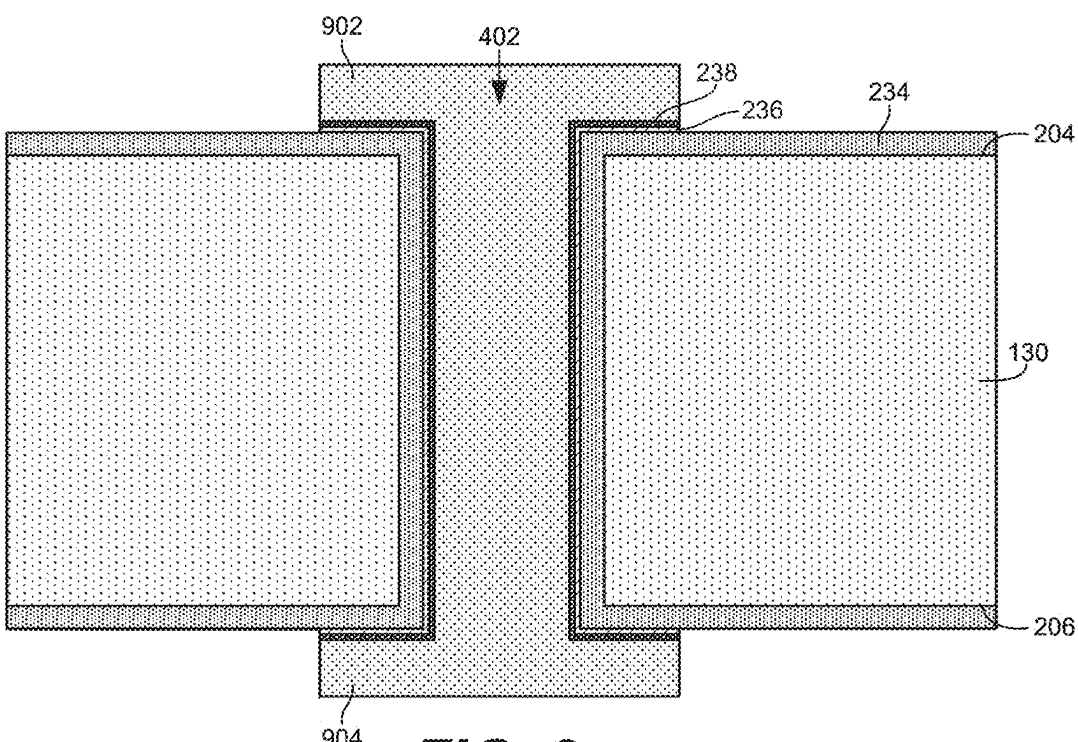


FIG. 9

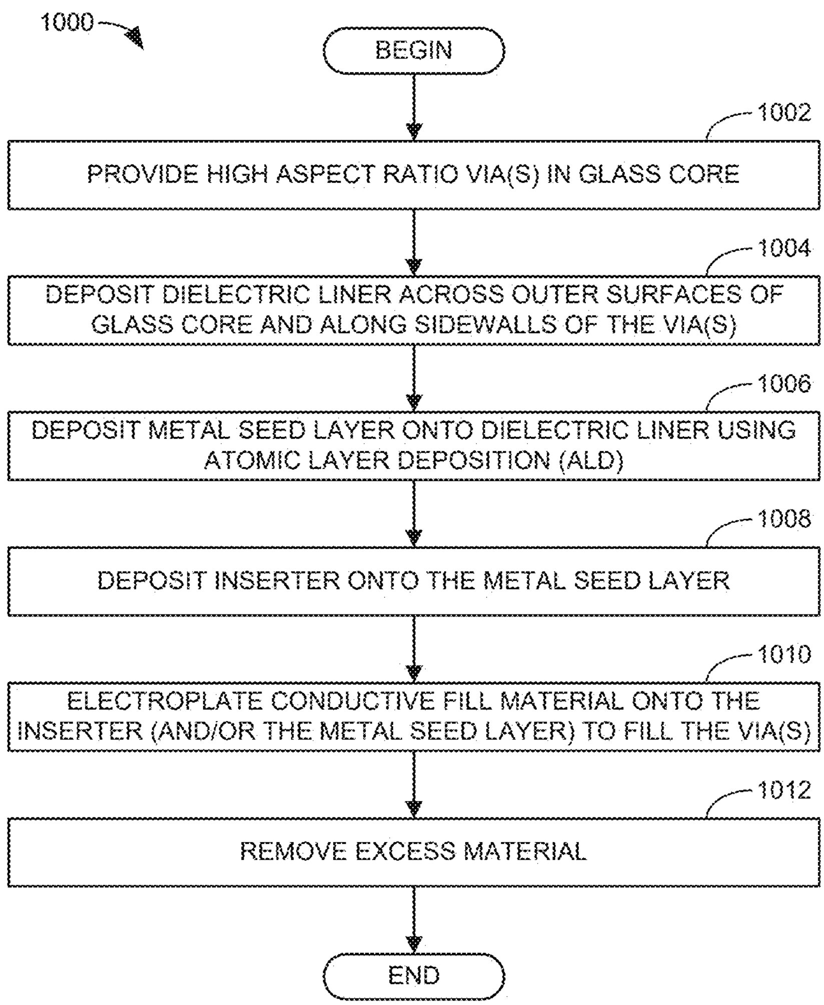

1000
BEGIN
1002
PROVIDE HIGH ASPECT RATIO VIA(S) IN GLASS CORE
1004
DEPOSIT DIELECTRIC LINER ACROSS OUTER SURFACES OF GLASS CORE AND ALONG SIDEWALLS OF THE VIA(S)
1006
DEPOSIT METAL SEED LAYER ONTO DIELECTRIC LINER USING ATOMIC LAYER DEPOSITION (ALD)
1008
DEPOSIT INSERTER ONTO THE METAL SEED LAYER
1010
ELECTROPLATE CONDUCTIVE FILL MATERIAL ONTO THE INSERTER (AND/OR THE METAL SEED LAYER) TO FILL THE VIA(S)
1012
REMOVE EXCESS MATERIAL
END

FIG. 10

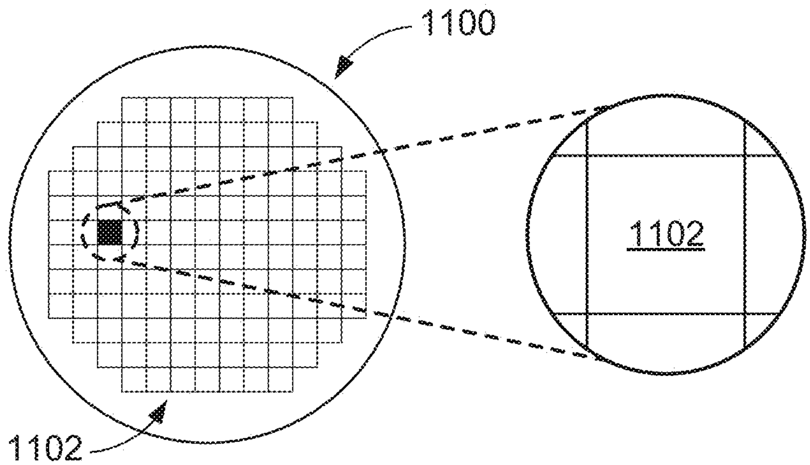

1100
1102
1102

FIG. 11

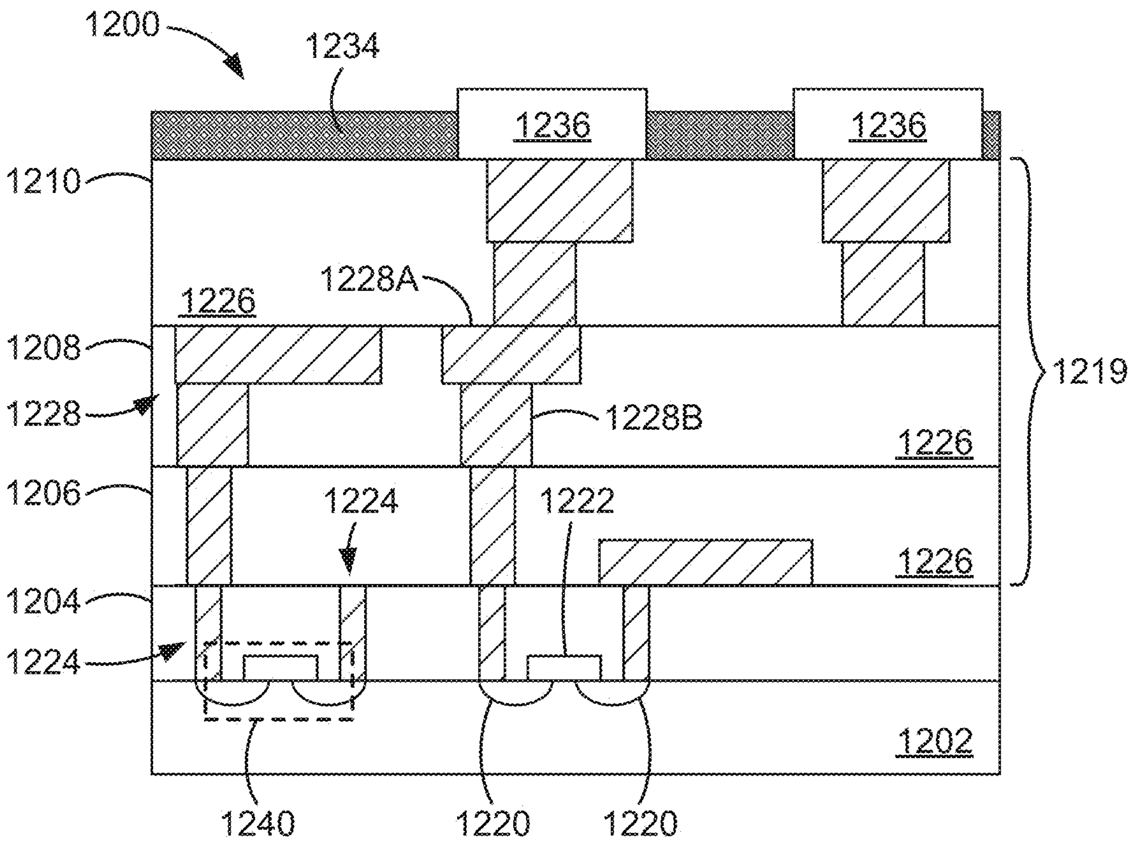

1200
1234
1236
1236
1210
1226
1228A
1208
1228
1228B
1226
1206
1224
1222
1226
1204
1224
1202
1219
1240
1220
1220

FIG. 12

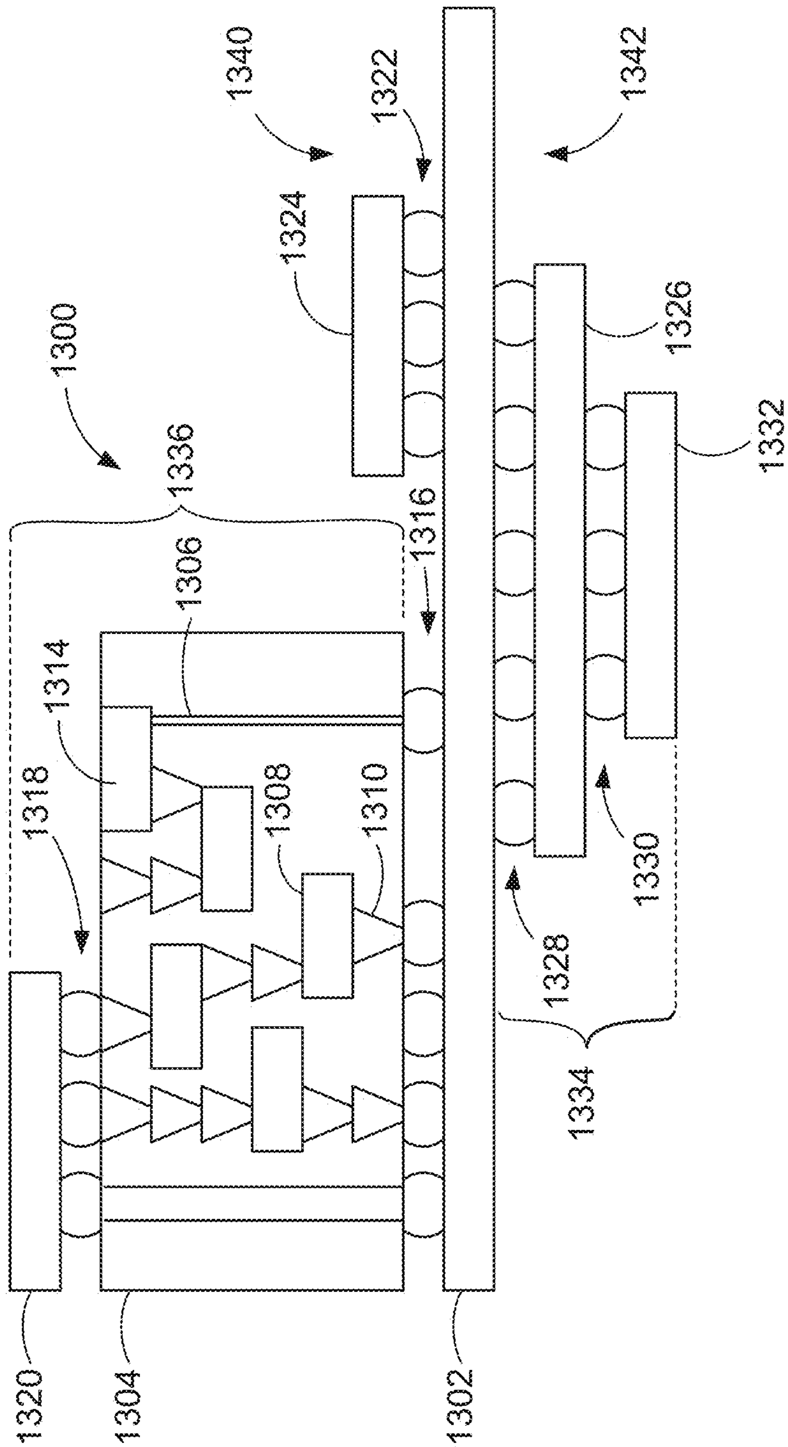

1300
1302
1304
1306
1308
1310
1314
1316
1318
1320
1322
1324
1326
1328
1330
1332
1334
1336
1340
1342

FIG. 13

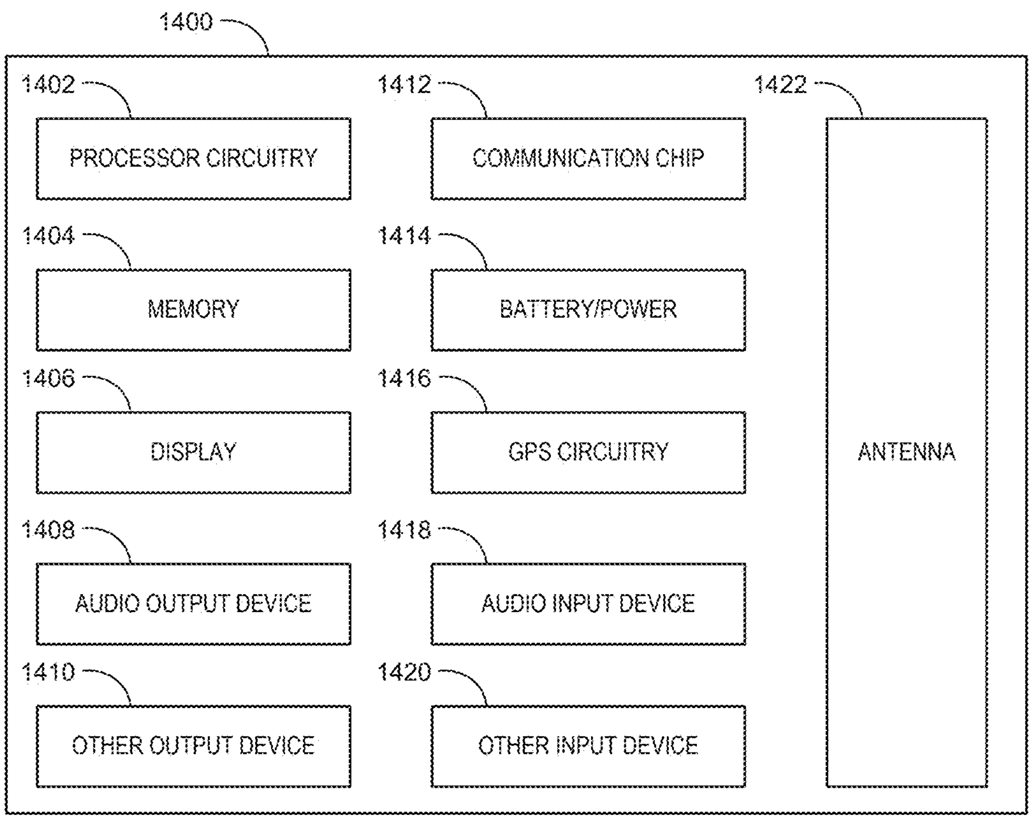

1400
1402
PROCESSOR CIRCUITRY
1412
COMMUNICATION CHIP
1422
1404
MEMORY
1414
BATTERY/POWER
1406
DISPLAY
1416
GPS CIRCUITRY
ANTENNA
1408
AUDIO OUTPUT DEVICE
1418
AUDIO INPUT DEVICE
1410
OTHER OUTPUT DEVICE
1420
OTHER INPUT DEVICE

FIG. 14

HIGH ASPECT RATIO VIAS WITH LOW STRESS IN INTEGRATED CIRCUIT PACKAGES

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to integrated circuit packages and, more particularly, to high aspect ratio vias with low stress in integrated circuit packages.

BACKGROUND

[0002] In many integrated circuit packages, one or more semiconductor dies are mechanically and electrically coupled to an underlying package substrate. In many cases, such package substrates include a package core that provides structural integrity to the package substrate. In recent years, package substrates have been developed using glass cores. By using laser-assisted etching, crack free, high-density vias (e.g., openings or holes) are formed into a glass core and subsequently plated with metal to serve as electrical pathways or interconnects to carry power and/or signals to and/or from a semiconductor die and/or other electrical component(s) on the package substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** illustrates an example integrated circuit (IC) package constructed in accordance with teachings disclosed herein.

[0004] FIG. **2** is a cross-sectional view of an example implementation of the package substrate of FIG. **1** at a point in the fabrication process in which the buildup regions of FIG. **1** are partially formed.

[0005] FIG. **3** is a cross-sectional view of another example implementation of the package substrate of FIG. **1**.

[0006] FIGS. **4-9** illustrate different stages during an example fabrication process to manufacture the example through-glass vias (TGVs) shown in FIGS. **2** and/or **3**.

[0007] FIG. **10** is a flowchart representative of an example method to manufacture any one of the example TGVs of FIGS. **2-9**.

[0008] FIG. **11** is a top view of a wafer including dies that may be included in an IC package constructed in accordance with teachings disclosed herein.

[0009] FIG. **12** is a cross-sectional side view of an IC device that may be included in an IC package constructed in accordance with teachings disclosed herein.

[0010] FIG. **13** is a cross-sectional side view of an IC device assembly that may include an IC package constructed in accordance with teachings disclosed herein.

[0011] FIG. **14** is a block diagram of an example electrical device that may include an IC package constructed in accordance with teachings disclosed herein.

[0012] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular.

DETAILED DESCRIPTION

[0013] Glass cores for package substrates provide mechanical and electrical benefits over organic cores (e.g., epoxy-based prepreg layer with glass cloth). Specifically, a glass core provides greater structural rigidity than a similarly sized organic core. Further, glass cores can enable smaller vias distributed at smaller pitches than is possible with organic cores, thereby enabling higher interconnect densities. However, a challenge with glass cores is the large difference in coefficient of thermal expansion (CTE) of the glass relative to the CTE of the conductive material (e.g., plated copper) used for the interconnects extending through the glass core. As a result, there is risk of introducing stress and possibly causing damage to a glass core from the plastic deformation (e.g., expansion and contraction) of the copper relative to the glass core during the different temperature cycles experienced during the fabrication of an associated integrated circuit package (e.g., high temperatures during reflow operations).

[0014] Examples disclosed herein mitigate against the concerns of a CTE mismatch by employing a seed layer stack of multiple different materials between the glass core and the copper subsequently plated thereon. More particularly, in some examples, the different materials in the seed layer stack include at least one of a stress reducing liner, a metal seed layer, and an inserter (any of which may themself contain one or more layers of materials). In some examples, the stress reducing liner is a low modulus material that absorbs some of the stress created by the CTE mismatch between the glass and plated copper. The metal seed layer provides electrical conductivity to enable electroplating of the copper. In some such examples, the inserter serves as a transition between the metal seed layer and the plated copper to facilitate the electroplating process and promote adhesion of the copper to the underlying metal seed layer.

[0015] Reducing stress between a glass core and plated vias extending through the glass core becomes particularly challenging as the vias reduce in width or diameter because of a resulting increase in the height to width aspect ratio of the vias. In particular, standard techniques for depositing a seed layer into a via (e.g., using physical vapor deposition (PVD)) are limited to aspect ratios of no more than 5 (e.g., vias 5 times longer (taller) than they are wide). Examples disclosed herein rely on different techniques, such as atomic layer deposition (ALD), to deposit the materials in the seed layer stack within vias having aspect ratios much greater than 5 (e.g., at least 6, at least 7, at least 9, at least 10, at least 12, at least 15, at least 20, etc.). ALD can be used to line the inner surfaces of vias with high aspect ratios because ALD enables conformal coatings regardless of the position and/or orientation of the surfaces being coated and enables high coverage across uneven surfaces. Furthermore, the thickness of films or coatings deposited using ALD can be precisely controlled (e.g., to the sub-nanometer range). While ALD can be employed to deposit materials into high-aspect ratio vias, not all materials are suitable for deposition using ALD techniques. Accordingly, examples disclosed herein identify suitable materials that serve the purposes of the different layers in the seed layer stack while also being capable of deposition through ALD. In this manner, stress in glass cores with high aspect ratio vias (e.g., through glass vias (TGVs)) can be significantly reduced to improve mechanical reliability of the core and increase yields. In particular, simulated experiments have shown examples disclosed herein achieve a reduction in stress in a glass core to less than one half (e.g., one third) the stress experienced in glass cores manufactured using known techniques.

[0016] FIG. **1** illustrates an example integrated circuit (IC) package **100** constructed in accordance with teachings disclosed herein. In the illustrated example, the IC package **100** is electrically coupled to a circuit board **102** via an array of contact pads or lands **104** on a mounting surface **105** (e.g., a bottom surface) of the package **100**. In some examples, the IC package **100** may include balls, pins, and/or pads, in addition to or instead of the contact pads **104**, to enable the electrical coupling of the package **100** to the circuit board **102**. In this example, the package **100** includes two semiconductor (e.g., silicon) dies **106**, **108** (sometimes also referred to as chips or chiplets) that are mounted to a package substrate **110** and enclosed by a package lid or mold compound **112**. Thus, the package substrate **110** is an example means for supporting a semiconductor die. While the example IC package **100** of FIG. **1** includes two dies **106**, **108**, in other examples, the package **100** may have only one die or more than two dies. In some examples, one of the dies **106**, **108** (or a separate die) is embedded in the package substrate **110**. The dies **106**, **108** can provide any suitable type of functionality (e.g., data processing, memory storage, etc.).

[0017] As shown in the illustrated example, each of the dies **106**, **108** is electrically and mechanically coupled to the substrate **110** via corresponding arrays of interconnects **114**. In FIG. **1**, the interconnects are shown as bumps. However, the interconnects **114** may be any other type of electrical connection in addition to or instead of the bumps shown (e.g., balls, pins, pads, wire bonding, etc.). The electrical connections between the dies **106**, **108** and the substrate **110** (e.g., the interconnects **114**) are sometimes referred to as first level interconnects. By contrast, the electrical connections between the IC package **100** and the circuit board **102** (e.g., the pads **104**) are sometimes referred to as second level interconnects. In some examples, the second level interconnects are used to electrically couple the IC package **100** to some component other than a circuit board (e.g., an interposer, another IC package, etc.). In some examples, one or both of the dies **106**, **108** may be stacked on top of one or more other dies and/or an interposer. In such examples, the dies **106**, **108** are coupled to the underlying die and/or interposer through a first set of first level interconnects and the underlying die and/or interposer may be connected to the package substrate **110** via a separate set of first level interconnects associated with the underlying die and/or interposer. Thus, as used herein, first level interconnects refer to interconnects (e.g., balls, bumps, pins, pads, wire bonding, etc.) between a die and a package substrate or a die and an underlying die and/or interposer.

[0018] As shown in FIG. **1**, the interconnects **114** of the first level interconnects include two different types of bumps corresponding to core bumps **116** and bridge bumps **118**. As used herein, the core bumps **116** are bumps on the dies **106**, **108** through which electrical signals pass between the dies **106**, **108** and components external to the IC package **100**. More particularly, as shown in the illustrated example, when the dies **106**, **108** are mounted to the package substrate **110**, the core bumps **116** are physically connected and electrically coupled to contact pads **120** on an inner surface **122** of the substrate **110**. The contact pads **120** on the inner surface **122** of the package substrate **110** are electrically coupled to the pads **104** on the bottom (external) surface **105** of the substrate **110** (e.g., a surface opposite the inner surface **122**) via internal interconnects **124** within the substrate **110**. As a result, there is a continuous electrical signal path (e.g., a continuous electrical signal path) between the interconnects **114** of the dies **106**, **108** and the pads **104** mounted to the circuit board **102** that pass through the contact pads **120** and the interconnects **124** provided therebetween.

[0019] As used herein, the bridge bumps **118** are bumps on the dies **106**, **108** through which electrical signals pass between different ones of the dies **106**, **108** within the package **100**. Thus, as shown in the illustrated example, the bridge bumps **118** of the first die **106** are electrically coupled to the bridge bumps **118** of the second die **108** via an interconnect bridge **126** embedded in the package substrate **110**. As represented in FIG. **1**, core bumps **116** are typically larger than bridge bumps **118**. In some examples, the interconnect bridge **126** and the associated bridge bumps **118** are omitted.

[0020] In some examples, an underfill material **119** is disposed between the dies **106**, **108** and the package substrate **110** around and/or between the first level interconnects **114** (e.g., around and/or between the core bumps **116** and/or the bridge bumps **118**). In the illustrated example, only the first die **106** is associated with the underfill material **119**. However, in other examples, both dies **106**, **108** are associated with the underfill material **119**. In other examples, the underfill material **119** is omitted. In some examples, the mold compound **112** is used as an underfill material that surrounds the first level interconnects **114**.

[0021] In some examples, the IC package **100** includes additional passive components, such as surface-mount resistors, capacitors, and/or inductors disposed the bottom (external) surface **105** of the package substrate **110** and/or the top (inner) surface **122** of the package substrate **110**.

[0022] For purposes of illustration, the internal interconnects **124** are shown as straight lines extending directly between the pads **104** on the bottom surface **105** and the contact pads on the inner surface **122**. However, in some examples, the internal interconnects **124** are defined by traces or routing in separate conductive (e.g., metal) layers within buildup regions **128** on one or both sides of a substrate core **130** (e.g., a base substrate) in the package substrate **110**. In such examples, the buildup regions **128** include dielectric layers to separate the different conductive layers. In such examples, the traces or routing in the different conductive layers are electrically coupled (to define the complete electrical path of the internal interconnects **124**) by conductive (e.g., metal) vias extending between the different conductive layers. Further, in some examples, the internal interconnects **124** include vias that extend through the substrate core **130**.

[0023] In some examples, the substrate core **130** is a glass substrate or glass core. In some examples, glass substrates (e.g., the glass core **130**) includes quartz, fused silica, and/or borosilicate glass. In some examples, glass substrates (e.g., the glass core **130**) includes at least 20% (by weight) of each of silicon (Si) and oxygen (O). In other examples, glass substrates (e.g., the glass core **130**) includes greater amounts of at least one of silicon or oxygen (e.g., at least 25 wt %, at least 30 wt %, at least 35 wt %, at least 40 wt %, etc.). In some examples, glass substrates (e.g., the glass core **130**) includes at least 5% (by weight) of aluminum (Al). In some examples, glass substrates (e.g., the glass core **130**) include at least one glass layer and do not include epoxy and do not include glass fibers (e.g., does not include an epoxy-based prepreg layer with glass cloth). In some examples, glass substrates (e.g., the glass core **130**) correspond to a single piece of glass that extends the full height/thickness of the core. In some examples, the glass core **130** has a rectangular shape that is substantially coextensive, in plan view, with the layers above and below the core (e.g., substantially coextensive with the buildup regions **128**). The glass core **130** provides stiffness and mechanical support or strength for the package substrate **110** and the rest of the package **100**. Thus, the glass core **130** is an example means for strengthening the package substrate **110**. In some examples, the thickness of the glass core **130** is driven by the size (e.g., footprint) of the package **100**. For example, in some instances, larger packages **100** include a substrate **110** with a larger (e.g., thicker) core **130** as compared with smaller packages **100** where the core does not need to be as thick.

[0024] FIG. **2** is a cross-sectional view of an example implementation **200** of the package substrate **110** of FIG. **1** at a point in the fabrication process in which the buildup regions **128** of FIG. **1** are partially formed. Specifically, the package substrate **110** shown in FIG. **2** includes a plurality of through-glass vias (TGVs) **202** (e.g., openings, through-holes, etc.) that extend all the way through the core **130** from a first (outer) surface **204** of the core **130** to a second (outer) surface **206** of the core. The TGVs **202** include a conductive fill material **208** (e.g., plated copper) that has been electroplated onto a seed layer stack **210** that lines walls **211** (sidewalls, inner surfaces) of the TGVs **202**. In some examples, the electroplating of the conductive fill material **208** is implemented to fill the TGVs **202**. That is, the conductive fill material **208** fills a central region of the TGVs **202**.

[0025] As shown in the illustrated example, the TGVs **202** are coupled to respective first via pads **212** that extend along the first surface **204** of the core **130** in a first metal layer **214** of the buildup region **128** on the first surface **204**. Likewise, the TGVs **202** are coupled to respective second via pads **216** that extend along the second surface **206** of the core **130** in a first metal layer **218** of the buildup region **128** on the second surface **206**. In some examples, via pads **212**, **216** include a portion of the seed layer stack **210** that extends along the outer surfaces **204**, **206** of the core **130**. In this example, a first dielectric layer **220** is deposited over the first via pads **212** and the first surface **204** of the core **130** and a second dielectric layer **222** is deposited over the second via pads **216** and the second surface **206** of the core **130**. Further, as shown in the illustrated example, additional metal vias **224** extend through respective ones of the dielectric layers **220**, **222**, to electrical couple the via pads in the first metal layers **214**, **218** adjacent the core **130** with second metal layers **226**, **228** on opposite sides of the dielectric layers **220**, **222**. In this example, the TGVs **202** and the associated additional vias **224** define at least portions of the internal interconnects **124** shown in FIG. **1**.

[0026] As represented in the illustrated example, the TGVs **202** have a height **230** (e.g., a length, a depth) that is significantly greater than a width **232** (e.g., a diameter) of the TGVs **202**. Thus, the TGVs **202** have a relatively high height-to-width aspect ratio. In this example, the height **230** of the TGVs corresponds to a thickness of the core **130**. As packages continue to increase in size, the thickness of the core **130** and, thus, the height **230** of the TGVs **202** also increases. In some examples, the core **130** thickness (and height **230** of the TGVs **202**) can be at least 800 micrometers (μm) or more (e.g., at least 900 μm, at least 1000 μm (1 millimeter (mm)), at least 1.2 mm, at least 1.4 mm, etc.). While core thicknesses (and thus TGV heights **230**) are increasing as technology advances, the width **232** of TGVs **202** have remained constant or have even become smaller as technology advances, thereby leading to ever increasing aspect ratios. For instance, the width **232** of TGVs **202** can be significantly less than or equal to approximately 100 μm (e.g., less than or equal to approximately 75 μm, less than or equal to approximately 55 μm, less than or equal to approximately 45 μm, less than or equal to approximately 30 μm, etc.). Thus, based on these dimensions, the aspect ratio of the TGVs **202** can be much greater than 5 (e.g., at least 6, at least 7, at least 9, at least 10, at least 12, at least 15, at least 20, etc.). Such large aspect ratios make it difficult if not impossible to line the walls **211** with materials for the seed layer stack **210** using standard deposition techniques. More particularly, the standard PVD process for depositing a metal seed layer is typically limited to aspect ratios of no more than 5.

[0027] One reason for the limitations of PVD processes is that PVD is a directional deposition process that deposits material unevenly depending on the orientation of the surface onto which material is being deposited as well as the position and/or distance of the surface relative to other surface onto which the material is being deposited. That is, PVD results in a non-conformal coating or film on an underlying substrate that differs in thickness depending on the location of the coating or film where the thickness is measured. For instance, when depositing a material onto a substrate that includes a through-hole (such as the TGVs **202** of FIG. **2**), far more material will accumulate on the surface of the substrate surrounding the opening of the through-hole with less and less material attaching to the wall of the through-hole the farther into the through-hole one goes. For at least these reasons, attempting to deposit a metal seed layer onto the walls **211** of the TGVs **202** (with high aspect ratios) are likely to result in gaps or voids in the coverage of the walls **211**, which may lead to gaps or voids in the conductive fill material **208** electroplated onto such seed layers.

[0028] The directional nature of PVD also means that the deposition of material onto an underlying surface requires direct line of sight. However, many TGVs **202** do not have perfectly straight walls **211**. Rather, as shown in the example TGV **300** shown schematically in FIG. **3**, TGVs **300** often include tapered (e.g., slanted, angled) walls that define overhanging features that prevent direct line of sight deposition. Therefore, the use of PVD is not a viable option to deposit a seed layer within a TGV that can then be used to fill the TGV with electroplated copper. Any one of the TGVs **202** of FIG. **2** may alternatively have a profile similar to the example TGV **300** of FIG. **3**. Further, in other examples, TGVs can have any other suitable profile.

[0029] Examples disclosed herein overcome the above limitations of PVD through the use of atomic layer deposition (ALD) techniques. Unlike PVD, ALD is not a directional deposition process. ALD provides a conformal coating onto all exterior surfaces of a substrate regardless of whether there is a direct line of sight. That is, unlike PVD, the thickness of a layer of material deposited using ALD can be substantially consistent across all surfaces onto which the material is deposited regardless of the position and/or orientation of the surfaces or their spatial relationship relative to one another. In other words, as shown in FIG. **3**, the seed layer **302** has a substantially even thickness with a first thickness **304** at a first end of the TGV **300** that is approximately equal to a second thickness of the seed layer **302** at a second end of the TGV **300**. Further, as shown in the illustrated example, the seed layer **302** has a third thickness **306** at a midpoint of the TGV **300** that is approximately equal to the first thickness **304** and approximately equal to the second thickness **306**. Further, each of the first, second, and third thicknesses **304**, **306**, **308** are approximately equal to a fourth thickness **310** of the seed layer **302** on the outer surfaces **204**, **206** of the core **130**. Stated more generally, in some examples, due to the use of ALD, the seed layer **302** has a thickness that differs by less than 5% between any two points on the seed layer **302**. As used herein, different thicknesses are "approximately equal" with there is less than 5% different between the thicknesses. Likewise, as used herein, a "substantially consistent thickness" and a "substantially even thickness" of a layer of material mean the layer has a thickness that varies by less than 5% across the entire area covered by the layer of material.

[0030] In some examples, as shown in the inset of FIG. **2**, the seed layer stack **210** includes a stack of multiple different layers of materials. In some examples, one or more of the different layers of materials in the seed layer stack **210** are deposited using an ALD process. In this example, the seed layer stack **210** includes a dielectric liner **234** (e.g., a stress reducing liner, a stress buffer layer), a metal seed layer **236** (e.g., a first conductive material, a conductive liner), and an inserter **238** (e.g., a second conductive material, a transition layer). More particularly, in this example, the metal seed layer **236** is between the dielectric liner **234** and the inserter **238**. Further, in this example, the dielectric liner **234** is closer to the core **130** than either the metal seed layer **236** or the inserter **238** is to the core **130**. Specifically, in this example, the dielectric liner **234** is in contact with the core **130** (e.g., in contact with the sidewalls **211** of the TGVs **202**). In some examples, the dielectric liner **234** is omitted such that the metal seed layer **236** is directly in contact with the core **130**. As shown in the illustrated example, the inserter **238** is closer to the conductive fill material **208** than either the dielectric liner **234** or the metal seed layer **236** is to the conductive fill material **208**. Specifically, in this example, the inserter **238** is in contact with the conductive fill material **208**. In some examples, the inserter **238** is omitted such that the metal seed layer **236** is in direct contact with the conductive fill material **208**.

[0031] In some examples, the dielectric liner **234** is a relatively low modulus material (e.g., materials with a Young's modulus of less than 2 Gigapascals (GPa)) that reduces stress resulting from the CTE mismatch between the glass core **130** and the metal used for the conductive fill material **208**. In some examples, the conductive fill material **208** includes copper. In other examples, a different metal can be used for the conductive fill material **208**. In some examples, the dielectric liner **234** includes a polymer (e.g., a parylene, polyimide, an epoxy), $Si_xO_yC_zH_w$, $SiN_xC_yH_z$, and/or SiO_x. In some examples, the dielectric liner **234** includes an organic material that contains an inorganic filler. In some examples, the dielectric liner **234** includes a fiber reinforced polymer. In some examples, the dielectric liner **234** includes multiple layers of different materials.

[0032] In some examples, the metal seed layer **236** includes an electrically conductive material to carry a current applied to the substrate (e.g., the core **130**) to facilitate the electroplating of the conductive fill material **208**. Further, in this example, the metal seed layer **236** includes materials capable of being deposited using ALD. More particularly, in some examples, the metal seed layer **236** includes at least one of ruthenium or copper and/or oxides of such metals. However, other materials may also be employed for the metal seed layer **236** (e.g., molybdenum, titanium, vanadium, chromium, iron, cobalt, nickel, zinc, zirconium, rhodium, palladium, silver, cadmium, tantalum, tungsten, iridium, platinum, gold, etc. and/or oxides of such metals). In some examples, the metal seed layer **236** includes any suitable electrically conductive metal oxide layer (e.g., zinc oxide). In some examples, the metal seed layer **236** does not include titanium or at least includes only trace amounts of titanium (e.g., less than 1 wt %, less than 0.5 wt %, less than 0.1 wt %, etc.). In some examples, the metal seed layer **236** includes multiple layers of different materials. In some examples, the metal seed layer **236** includes different materials than the conductive fill material **208**. That is, in some examples, the metal seed layer **236** includes a different composition from the conductive fill material **208**.

[0033] In this example, the inserter **238** serves as a transition layer between the metal seed layer **236** and the conductive fill material **208** to improve adhesion between the metal seed layer **236** and the conductive fill material **208**. In some examples, the inserter **238** includes conductive materials capable of being deposited using ALD. In some examples, the inserter **238** includes at least one of titanium or copper. In some examples, the inserter **238** includes at least one of TiN, SiN, TaN, or SiOx. In some examples, the inserter **238** includes multiple layers of different materials.

[0034] In some examples, each of the layers of materials in the seed layer stack **210** have different thicknesses. More particularly, in some examples, the dielectric liner **234** has a first thickness **240** that is significantly greater than a second thickness **242** of the metal seed layer **236** and significantly greater than a third thickness **244** of the inserter **238**. The greater thickness of the dielectric liner **234** is to provide stress relief as discussed above. In some examples, the first thickness **240** of the dielectric liner is at least 300 nanometers (nm) (0.3 μm) and may be significantly thicker (e.g., at least 0.5 μm, at least 0.75 μm, at least 1 μm, at least 2 μm, at least 5 μm, at least 7.5μm, at least 9 μm, etc.). By contrast, in some examples, the second thickness **242** of the metal seed layer **236** is less than or equal to approximately 100 nm (e.g., less than or equal to approximately 75 nm, less than or equal to approximately 50 nm, less than or equal to approximately 25 nm, less than or equal to approximately 10 nm, less than or equal to approximately 5 nm, less than or equal to approximately 3 nm, etc.) and may be as thin as approximately 1 nm or less. In some examples, the second thickness **242** is greater than 100 nm. In some examples, the third thickness **244** of the inserter **238** is also less than or equal to approximately 100 nm (e.g., less than or equal to approximately 75 nm, less than or equal to approximately 50 nm, less than or equal to approximately 25 nm, less than or equal to approximately 10 nm, less than or equal to approximately 5 nm, less than or equal to approximately 3 nm, etc.) and may be as thin as approximately 1 nm or less. In some examples, the third thickness **244** is greater than 100 nm. In some examples, the third thickness **244** of the inserter **238** is less than or equal to the second thickness **242** of the metal seed layer **236**. In other examples, the third thickness **244** of the inserter **238** is greater than the second thickness **242** of the metal seed layer **236**.

[0035] FIGS. **4-9** illustrate different stages during an example fabrication process to manufacture a plated through-hole or TGV such as the TGVs **202**, **300** shown in FIGS. **2** and/or **3**. More particularly, FIG. **4** represents the stage of fabrication following the creation of a via **402** (e.g., a TGV, a through-hole, an opening) extending through the glass core **130**. In this example, the TGV **402** is defined by an inner sidewall **404** that extends in a straight line from the first and second (outer) surfaces **204**, **206** of the core **130**. More particularly, in this example the sidewall **404** extends in a direction substantially perpendicular (e.g., within 5 degrees of perpendicular) to the outer surfaces **204**, **206**. However, in other examples, the sidewall **404** may extend at non-perpendicular angles relative to the outer surfaces **204**, **206**. Further, in some examples, the sidewall **404** may not be a continuously straight line. Rather, in some examples, the sidewall **404** may define an hour-glass shape, such as is shown in the illustrated example of FIG. **3**. In other examples, the sidewall **404** can have any other suitable shape or profile.

[0036] FIG. **5** represents the stage of fabrication following the deposition of the dielectric liner **234** onto exposed surfaces of the core **130**. In this example, the exposed surfaces include the outer surfaces **204**, **206** of the core **130** as well as the sidewall **404** of the TGV **402** in the core **130**. In some examples, the dielectric liner **234** is deposited via spin coating, slit coating, chemical vapor deposition (CVD), or vacuum lamination. In some examples, these deposition processes result in the TGV **402** being substantially or completely filled by the dielectric liner **234**. In such examples, a subsequent laser drilling process through the filled TGV **402** is performed to remove the dielectric liner **234** from the central region of the TGV **402** so that the dielectric liner **234** is limited to (e.g., lines) the sidewall **404** of the TGV **402**.

[0037] FIG. **6** represents a subsequent stage of fabrication after the deposition of the metal seed layer **236** onto the dielectric liner **234**. In this example, because the TGV **402** has a relatively high aspect ratio (e.g., above **5**), standard PVD processes are unavailable. Accordingly, in some examples, the metal seed layer **236** is deposited through an ALD process. More particularly, in some examples, the metal seed layer **236** is deposited using thermal ALD or plasma enhanced ALD.

[0038] FIG. **7** represents a subsequent stage of fabrication after the deposition of the inserter **238** onto the metal seed layer **236**. In some examples, the deposition process employed to deposit the inserter **238** may depend on the material used for the inserter **238**. For instance, if the inserter **238** includes a titanium/copper layer, the inserter can be deposited via sputtering. In other examples, if the inserter **238** is to be a dry deposited seed metal (e.g., copper or other suitable metal), an ALD process can be used. As discussed above, the dielectric liner **234**, the metal seed layer **236**, and the inserter **238** collectively correspond to the seed layer stack **210** discussed above in connection with FIGS. **2** and **3**. In some examples, the inserter **238** is omitted.

[0039] FIG. **8** represents a subsequent stage of fabrication after the deposition of the conductive fill material **208** onto the metal seed layer **236** (and the inserter **238**). In some examples, the conductive fill material **208** is deposited via electroplating using the metal seed layer **236** (and inserter **238**) as the conductive underlying surface. As shown in the illustrated example, the conductive fill material **208** is electroplated to fill a central region of (e.g., the remaining open space within) the TGV **402**. Further, as shown in FIG. **8**, the conductive fill material **208** also extends across the outer surfaces **204**, **260** of the core **130**.

[0040] FIG. **9** represents a subsequent stage of fabrication after the removal of excess portions of the conductive fill material **208**, the inserter **238**, and/or the metal seed layer **236** on the outer surfaces **204**, **206** of the core **130**. In some examples, the excess portions of material are removed via an etching process (e.g., a wet etch, a dry etch, etc.). As shown in the illustrated example of FIG. **9**, the remaining portions of the material on the outer surfaces **204**, **206** define first and second via pads **902**, **904** at either end of the TGV **402**.

[0041] FIG. **10** is a flowchart representative of an example method **1000** to manufacture any one of the example TGVs **202**, **300**, **402** of FIGS. **2-9**. In some examples, some or all of the operations outlined in the example method of FIG. **10** are performed automatically by fabrication equipment that is programmed to perform such operations. Although the example method of manufacture is described with reference to the flowchart illustrated in FIG. **10**, many other methods may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, in some examples, additional processing operations can be performed before, between, and/or after any of the blocks represented in the illustrated example.

[0042] Turning to FIG. **10** in detail, the example process begins at block **1002** by providing high aspect ratio via(s) (e.g., the vias **202**, **300**, **402**) in a glass core (e.g., the core **130**). The operation(s) associated with block **1002** are represented by FIG. **4** as described above. At block **1004**, the example process involves depositing a dielectric liner (e.g., the dielectric liner **234**) across outer surfaces (e.g., the surfaces **204**, **206**) of the glass core **130** and along sidewalls (e.g., the sidewalls **211**, **404**) of the via(s) **202**, **300**, **402**. The dielectric liner **234** serves as a stress buffer between the glass core **130** and the subsequently added metal layers. The operation(s) associated with block **1004** are represented by FIG. **5** as described above.

[0043] At block **1006**, the example process involves depositing a metal seed layer (e.g., the metal seed layer **236**) onto the dielectric liner **234** using atomic layer deposition (ALD). In this example, ALD is employed because of the high aspect ratio of the via(s) **202**, **300**, **402**. The operation(s) associated with block **1006** are represented by FIG. **6** as described above. At block **1008**, the example process involves depositing an inserter (e.g., the inserter **238**) onto the metal seed layer **236**. In some examples, block **1008** is omitted. The operation(s) associated with block **1008** are represented by FIG. **7** as described above. At block **1010**, the example process involves electroplating a conductive fill material (e.g., the conductive fill material **208**) onto the inserter **238** (and/or the metal seed layer **236**) to fill the via(s) **202**, **300**, **402**. The operation(s) associated with block **1010** are represented by FIG. **8** as described above. At block **1012**, the example process involves removing excess material. Specifically, portions of the conductive fill material **208**, the inserter **238**, and/or the metal seed layer **236** are removed (e.g., via etching) to define the first metal layers (e.g., the first metal layers **214**, **218**) on the outer surfaces **204**, **206** of the glass core **130**. The operation(s) associated with block **1012** are represented by FIG. **9** as described above. Thereafter, the example process ends, and the resulting apparatus may proceed to subsequent fabrication processes to, for example, add the buildup regions **128** onto the glass core **130**.

[0044] The example TGVs **202**, **300**, **402** disclosed herein may be included in any suitable electronic component. FIGS. **11-14** illustrate various examples of apparatus that may include and/or be included in the example IC package **100** of FIG. **1** that contains the example TGVs **202**, **300**, **402** disclosed herein.

[0045] FIG. **11** is a top view of a wafer **1100** and dies **1102** that may be included in the IC package **100** of FIG. **1** (e.g., as any suitable ones of the dies **106**, **108**) with a substrate that includes one or more of the example TGVs **202**, **300**, **402** disclosed herein. The wafer **1100** includes semiconductor material and one or more dies **1102** having circuitry. Each of the dies **1102** may be a repeating unit of a semiconductor product. After the fabrication of the semiconductor product is complete, the wafer **1100** may undergo a singulation process in which the dies **1102** are separated from one another to provide discrete "chips." The die **1102** includes one or more transistors (e.g., some of the transistors **1240** of FIG. **12**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., traces, resistors, capacitors, inductors, and/or other circuitry), and/or any other components. In some examples, the die **1102** may include and/or implement a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuitry or electronics. Multiple ones of these devices may be combined on a single die **1102**. For example, a memory array of multiple memory circuits may be formed on a same die **1102** as programmable circuitry (e.g., the processor circuitry **1402** of FIG. **14**) and/or other logic circuitry. Such memory may store information for use by the programmable circuitry. The example IC package **100** disclosed herein may be manufactured using a die-to-wafer assembly technique in which some dies are attached to a wafer **1100** that includes others of the dies, and the wafer **1100** is subsequently singulated.

[0046] FIG. **12** is a cross-sectional side view of an IC device **1200** that may be included in the example IC package **100** (e.g., in any one of the dies **106**, **108**) with a substrate that includes one or more of the example TGVs **202**, **300**, **402** disclosed herein. One or more of the IC devices **1200** may be included in one or more dies **1102** (FIG. **11**). The IC device **1200** may be formed on a die substrate **1202** (e.g., the wafer **1100** of FIG. **11**) and may be included in a die (e.g., the die **1102** of FIG. **11**). The die substrate **1202** may be a semiconductor substrate including semiconductor materials including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **1202** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some examples, the die substrate **1202** may be formed using alternative materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1202**. Although a few examples of materials from which the die substrate **1202** may be formed are described here, any material that may serve as a foundation for an IC device **1200** may be used. The die substrate **1202** may be part of a singulated die (e.g., the dies **1102** of FIG. **11**) or a wafer (e.g., the wafer **1100** of FIG. **11**).

[0047] The IC device **1200** may include one or more device layers **1204** disposed on and/or above the die substrate **1202**. The device layer **1204** may include features of one or more transistors **1240** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1202**. The device layer **1204** may include, for example, one or more source and/or drain (S/D) regions **1220**, a gate **1222** to control current flow between the S/D regions **1220**, and one or more S/D contacts **1224** to route electrical signals to/from the S/D regions **1220**. The transistors **1240** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1240** are not limited to the type and configuration depicted in FIG. **12** and may include a wide variety of other types and/or configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

[0048] Each transistor **1240** may include a gate **1222** including a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and/or zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and/or lead zinc niobate. In some examples, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

[0049] The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1240** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may include a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and/or any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and/or aluminum carbide), and/or any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

[0050] In some examples, when viewed as a cross-section of the transistor **1240** along the source-channel-drain direction, the gate electrode may include a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1202** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1202**. In other examples, at least one of the metal layers that form the gate electrode may be a planar layer that is substantially parallel to the top surface of the die substrate **1202** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1202**. In other examples, the gate electrode may include a combination of U-shaped structures and/or planar, non-U-shaped structures. For example, the gate electrode may include one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

[0051] In some examples, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and/or silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process operations. In some examples, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

[0052] The S/D regions **1220** may be formed within the die substrate **1202** adjacent to the gate **1222** of corresponding transistor(s) **1240**. The S/D regions **1220** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1202** to form the S/D regions **1220**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1202** may follow the ion-implantation process. In the latter process, the die substrate **1202** may first be etched to form recesses at the locations of the S/D regions **1220**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1220**. In some implementations, the S/D regions **1220** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some examples, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some examples, the S/D regions **1220** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further examples, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1220**.

[0053] Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1240**) of the device layer **1204** through one or more interconnect layers disposed on the device layer **1204** (illustrated in FIG. **12** as interconnect layers **1206-1210**). For example, electrically conductive features of the device layer **1204** (e.g., the gate **1222** and the S/D contacts **1224**) may be electrically coupled with the interconnect structures **1228** of the interconnect layers **1206-1210**. The one or more interconnect layers **1206-1210** may form a metallization stack (also referred to as an "ILD stack") **1219** of the IC device **1200**.

[0054] The interconnect structures **1228** may be arranged within the interconnect layers **1206-1210** to route electrical signals according to a wide variety of designs (in particular, the arrangement is not limited to the particular configuration of interconnect structures **1228** depicted in FIG. **12**). Although a particular number of interconnect layers **1206-1210** is depicted in FIG. **12**, examples of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

[0055] In some examples, the interconnect structures **1228** may include lines **1228*a*** and/or vias **1228*b*** filled with an electrically conductive material such as a metal. The lines **1228*a*** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1202** upon which the device layer **1204** is formed. For example, the lines **1228*a*** may route electrical signals in a direction in and/or out of the page from the perspective of FIG. **12**. The vias **1228*b*** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1202** upon which the device layer **1204** is formed. In some examples, the vias **1228*b*** may electrically couple lines **1228*a*** of different interconnect layers **1206-1210** together.

[0056] The interconnect layers **1206-1210** may include a dielectric material **1226** disposed between the interconnect structures **1228**, as shown in FIG. **12**. In some examples, the dielectric material **1226** disposed between the interconnect structures **1228** in different ones of the interconnect layers **1206-1210** may have different compositions; in other examples, the composition of the dielectric material **1226** between different interconnect layers **1206-1210** may be the same.

[0057] A first interconnect layer **1206** (referred to as Metal 1 or "M1") may be formed directly on the device layer **1204**. In some examples, the first interconnect layer **1206** may include lines **1228*a*** and/or vias **1228*b***, as shown. The lines **1228*a*** of the first interconnect layer **1206** may be coupled with contacts (e.g., the S/D contacts **1224**) of the device layer **1204**.

[0058] A second interconnect layer **1208** (referred to as Metal 2 or "M2") may be formed directly on the first interconnect layer **1206**. In some examples, the second interconnect layer **1208** may include vias **1228*b*** to couple the lines **1228*a*** of the second interconnect layer **1208** with the lines **1228*a*** of the first interconnect layer **1206**. Although the lines **1228*a*** and the vias **1228*b*** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1208**) for the sake of clarity, the lines **1228*a*** and the vias **1228*b*** may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some examples.

[0059] A third interconnect layer **1210** (referred to as Metal 3 or "M3") (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1208** according to similar techniques and/or configurations described in connection with the second interconnect layer **1208** or the first interconnect layer **1206**. In some examples, the interconnect layers that are "higher up" in the metallization stack **1219** in the IC device **1200** (i.e., further away from the device layer **1204**) may be thicker.

[0060] The IC device **1200** may include a solder resist material **1234** (e.g., polyimide or similar material) and one or more conductive contacts **1236** formed on the interconnect layers **1206-1210**. In FIG. **12**, the conductive contacts **1236** are illustrated as taking the form of bond pads. The conductive contacts **1236** may be electrically coupled with the interconnect structures **1228** and configured to route the electrical signals of the transistor(s) **1240** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1236** to mechanically and/or electrically couple a chip including the IC device **1200** with another component (e.g., a circuit board). The IC device **1200** may include additional or alternate structures to route the electrical signals from the interconnect layers **1206-1210**; for example, the conductive contacts **1236** may include other analogous features (e.g., posts) that route the electrical signals to external components.

[0061] FIG. **13** is a cross-sectional side view of an IC device assembly **1300** that may include the example IC package **100** of FIG. **1** with a substrate that includes one or more of the example TGVs **202**, **300**, **402** disclosed herein. In some examples, the IC device assembly corresponds to the example IC package **100** of FIG. **1**. The IC device assembly **1300** includes a number of components disposed on a circuit board **1302** (which may be, for example, a motherboard). The IC device assembly **1300** includes components disposed on a first face **1340** of the circuit board **1302** and an opposing second face **1342** of the circuit board **1302**; generally, components may be disposed on one or both faces **1340** and **1342**. Any of the IC packages discussed below with reference to the IC device assembly **2200** may take the form of the example IC package **100** of FIG. **1**.

[0062] In some examples, the circuit board **1302** may be a printed circuit board (PCB) including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1302**. In other examples, the circuit board **1302** may be a non-PCB substrate.

[0063] The IC device assembly **1300** illustrated in FIG. **13** includes a package-on-interposer structure **1336** coupled to the first face **1340** of the circuit board **1302** by coupling components **1316**. The coupling components **1316** may electrically and mechanically couple the package-on-interposer structure **1336** to the circuit board **1302**, and may include solder balls (as shown in FIG. **13**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

[0064] The package-on-interposer structure **1336** may include an IC package **1320** coupled to an interposer **1304** by coupling components **1318**. The coupling components **1318** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1316**. Although a single IC package **1320** is shown in FIG. **13**, multiple IC packages may be coupled to the interposer **1304**; indeed, additional interposers may be coupled to the interposer **1304**. The interposer **1304** may provide an intervening substrate used to bridge the circuit board **1302** and the IC package **1320**. The IC package **1320** may be or include, for example, a die (the die **1102** of FIG. **11**), an IC device (e.g., the IC device **1200** of FIG. **12**), or any other suitable component. Generally, the interposer **1304** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1304** may couple the IC package **1320** (e.g., a die) to a set of BGA conductive contacts of the coupling components **1316** for coupling to the circuit board **1302**. In the example illustrated in FIG. **13**, the IC package **1320** and the circuit board **1302** are attached to opposing sides of the interposer **1304**; in other examples, the IC package **1320** and the circuit board **1302** may be attached to a same side of the interposer **1304**. In some examples, three or more components may be interconnected by way of the interposer **1304**.

[0065] In some examples, the interposer **1304** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some examples, the interposer **1304** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some examples, the interposer **1304** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1304** may include metal interconnects **1308** and vias **1310**, including but not limited to through-silicon vias (TSVs) **1306**. The interposer **1304** may further include embedded devices **1314**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1304**. The package-on-interposer structure **1336** may take the form of any of the package-on-interposer structures known in the art.

[0066] The IC device assembly **1300** may include an IC package **1324** coupled to the first face **1340** of the circuit board **1302** by coupling components **1322**. The coupling components **1322** may take the form of any of the examples discussed above with reference to the coupling components **1316**, and the IC package **1324** may take the form of any of the examples discussed above with reference to the IC package **1320**.

[0067] The IC device assembly **1300** illustrated in FIG. **13** includes a package-on-package structure **1334** coupled to the second face **1342** of the circuit board **1302** by coupling components **1328**. The package-on-package structure **1334** may include a first IC package **1326** and a second IC package **1332** coupled together by coupling components **1330** such that the first IC package **1326** is disposed between the circuit board **1302** and the second IC package **1332**. The coupling components **1328**, **1330** may take the form of any of the examples of the coupling components **1316** discussed above, and the IC packages **1326**, **1332** may take the form of any of the examples of the IC package **1320** discussed above. The package-on-package structure **1334** may be configured in accordance with any of the package-on-package structures known in the art.

[0068] FIG. **14** is a block diagram of an example electrical device **1400** that may include one or more of the example IC package **100** of FIG. **1** with a substrate that includes one or more of the example TGVs **202**, **300**, **402** disclosed herein. For example, any suitable ones of the components of the electrical device **1400** may include one or more of the device assemblies **1300**, IC devices **1200**, or dies **1102** disclosed herein, and may be arranged in the example IC package **100**. A number of components are illustrated in FIG. **14** as included in the electrical device **1400**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some examples, some or all of the components included in the electrical device **1400** may be attached to one or more motherboards. In some examples, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

[0069] Additionally, in various examples, the electrical device **1400** may not include one or more of the components illustrated in FIG. **14**, but the electrical device **1400** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1400** may not include a display **1406**, but may include display interface circuitry (e.g., a connector and driver circuitry) to which a display **1406** may be coupled. In another set of examples, the electrical device **1400** may not include an audio input device **1418** (e.g., microphone) or an audio output device **1408** (e.g., a speaker, a headset, earbuds, etc.), but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1418** or audio output device **1408** may be coupled.

[0070] The electrical device **1400** may include programmable circuitry **1402** (e.g., one or more processing devices). The programmable circuitry **1402** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1400** may include a memory **1404**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some examples, the memory **1404** may include memory that shares a die with the programmable circuitry **1402**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

[0071] In some examples, the electrical device **1400** may include a communication chip **1412** (e.g., one or more communication chips). For example, the communication chip **1412** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1400**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some examples they might not.

[0072] The communication chip **1412** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-1205 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1412** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1412** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1412** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1412** may operate in accordance with other wireless protocols in other examples. The electrical device **1400** may include an antenna **1422** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

[0073] In some examples, the communication chip **1412** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1412** may include multiple communication chips. For instance, a first communication chip **1412** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1412** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some examples, a first communication chip **1412** may be dedicated to wireless communications, and a second communication chip **1412** may be dedicated to wired communications.

[0074] The electrical device **1400** may include battery/power circuitry **1414**. The battery/power circuitry **1414** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1400** to an energy source separate from the electrical device **1400** (e.g., AC line power).

[0075] The electrical device **1400** may include a display **1406** (or corresponding interface circuitry, as discussed above). The display **1406** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

[0076] The electrical device **1400** may include an audio output device **1408** (or corresponding interface circuitry, as discussed above). The audio output device **1408** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

[0077] The electrical device **1400** may include an audio input device **1418** (or corresponding interface circuitry, as discussed above). The audio input device **1418** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

[0078] The electrical device **1400** may include GPS circuitry **1416**. The GPS circuitry **1416** may be in communication with a satellite-based system and may receive a location of the electrical device **1400**, as known in the art.

[0079] The electrical device **1400** may include any other output device **1410** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1410** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

[0080] The electrical device **1400** may include any other input device **1420** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1420** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

[0081] The electrical device **1400** may have any desired form factor, such as a hand-held or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable electrical device. In some examples, the electrical device **1400** may be any other electronic device that processes data.

[0082] "Including" and "comprising" (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of "include" or "comprise" (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase "at least" is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term "comprising" and "including" are open ended. The term "and/or" when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0083] As used herein, singular references (e.g., "a", "an", "first", "second", etc.) do not exclude a plurality. The term "a" or "an" object, as used herein, refers to one or more of that object. The terms "a" (or "an"), "one or more", and "at least one" are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0084] As used herein, unless otherwise stated, the term "above" describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is "below" a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0085] Notwithstanding the foregoing, in the case of referencing a semiconductor device (e.g., a transistor), a semiconductor die containing a semiconductor device, and/or an integrated circuit (IC) package containing a semiconductor die during fabrication or manufacturing, "above" is not with reference to Earth, but instead is with reference to an underlying substrate on which relevant components are fabricated, assembled, mounted, supported, or otherwise provided. Thus, as used herein and unless otherwise stated or implied from the context, a first component within a semiconductor die (e.g., a transistor or other semiconductor device) is "above" a second component within the semiconductor die when the first component is farther away from a substrate (e.g., a semiconductor wafer) during fabrication/manufacturing than the second component on which the two components are fabricated or otherwise provided. Similarly, unless otherwise stated or implied from the context, a first component within an IC package (e.g., a semiconductor die) is "above" a second component within the IC package during fabrication when the first component is farther away from a printed circuit board (PCB) to which the IC package is to be mounted or attached. It is to be understood that semiconductor devices are often used in orientation different than their orientation during fabrication. Thus, when referring to a semiconductor device (e.g., a transistor), a semiconductor die containing a semiconductor device, and/or an integrated circuit (IC) package containing a semiconductor die during use, the definition of "above" in the preceding paragraph (i.e., the term "above" describes the relationship of two parts relative to Earth) will likely govern based on the usage context.

[0086] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0087] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in "contact" with another part is defined to mean that there is no intermediate part between the two parts.

[0088] Unless specifically stated otherwise, descriptors such as "first," "second," "third," etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor "first" may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as "second" or "third." In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0089] As used herein, "approximately" and "about" modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, "approximately" and "about" may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For example, "approximately" and "about" may indicate such dimensions may be within a tolerance range of +/−10% unless otherwise specified herein.

[0090] As used herein "substantially real time" refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, "substantially real time" refers to real time +1 second.

[0091] As used herein, the phrase "in communication," including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0092] As used herein, "programmable circuitry" is defined to include (i) one or more special purpose electrical circuits (e.g., an application specific circuit (ASIC)) structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmable with instructions to perform specific functions(s) and/or operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of programmable circuitry include programmable microprocessors such as Central Processor Units (CPUs) that may execute first instructions to perform one or more operations and/or functions, Field Programmable Gate Arrays (FPGAs) that may be programmed with second instructions to cause configuration and/or structuring of the FPGAs to instantiate one or more operations and/or functions corresponding to the first instructions, Graphics Processor Units (GPUs) that may execute first instructions to perform one or more operations and/or functions, Digital Signal Processors (DSPs) that may execute first instructions to perform one or more operations and/or functions, XPUs, Network Processing Units (NPUs) one or more microcontrollers that may execute first instructions to perform one or more operations and/or functions and/or integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of programmable circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more NPUs, one or more DSPs, etc., and/or any combination(s) thereof), and orchestration technology (e.g., application programming interface (s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of programmable circuitry is/are suited and available to perform the computing task(s).

[0093] As used herein integrated circuit/circuitry is defined as one or more semiconductor packages containing one or more circuit elements such as transistors, capacitors, inductors, resistors, current paths, diodes, etc. For example an integrated circuit may be implemented as one or more of an ASIC, an FPGA, a chip, a microchip, programmable circuitry, a semiconductor substrate coupling multiple circuit elements, a system on chip (SoC), etc.

[0094] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that enable the fabrication of high aspect ratio plated vias within a glass core. Plating metal (e.g., copper) within through-glass vias creates the risk of failures due to high stress imposed on the glass core as a result of the CTE mismatch between glass and the plated metal. However, examples disclosed herein mitigate this concern through the use of a dielectric liner between the glass core and the plated metal that serves as a stress buffer.

[0095] A further challenge with electroplating high aspect ratio vias is the inability to obtain full coverage of the inner sidewalls of the vias using standard deposition techniques. Examples disclosed herein overcome these challenges by employing atomic layer deposition (ALD) to deposit a metal seed layer within the vias. Not all conductive materials suitable for a metal seed layer are compatible with ALD and at least some of the conductive materials that are compatible do not provide strong adhesion with the metal to be subsequently electroplated thereto. Accordingly, in some examples, an inserter or transition layer is deposited onto the metal seed layer deposited using ALD to promote greater adhesion for the product of reliable high aspect ratio vias.

[0096] Further examples and combinations thereof include the following:

[0097] Example 1 includes a package substrate comprising a core having a via extending therethrough, a first conductive material in the via, a dielectric material at least partially between a wall of the via and the first conductive material, and a second conductive material in the via, the second conductive material closer to a central region of the via than the first conductive material is to the central region of the via, the first conductive material different from the second conductive material, the first conductive material at least partially between the dielectric material and the second conductive material.

[0098] Example 2 includes the package substrate of example 1, wherein the core is a glass core.

[0099] Example 3 includes the package substrate of any one of examples 1 or 2, wherein the first conductive material includes ruthenium, and the second conductive material includes copper.

[0100] Example 4 includes the package substrate of example 3, wherein the first conductive material includes both ruthenium and copper.

[0101] Example 5 includes the package substrate of any one of examples 1-4, wherein the via has an aspect ratio of at least 6.

[0102] Example 6 includes the package substrate of example 5, wherein the first conductive material has a first thickness at a first end of the via, a second thickness at a second end of the via, and a third thickness at a midpoint of the via, the first thickness approximately equal to the second thickness and approximately equal to the third thickness.

[0103] Example 7 includes the package substrate of any one of examples 1-6, wherein the dielectric material has a thickness between approximately 0.5 micrometers and 9 micrometers.

[0104] Example 8 includes the package substrate of any one of examples 1-7, wherein the first conductive material has a thickness less than 70 nanometers.

[0105] Example 9 includes the package substrate of any one of examples 1-8, further including a third conductive material at least partially between the first conductive material and the second conductive material, the third conductive material different from the first conductive material and different from the second conductive material.

[0106] Example 10 includes the package substrate of example 9, wherein the third conductive material has a thickness between approximately 5 nanometers and 100 nanometers.

[0107] Example 11 includes the package substrate of any one of examples 9 or 10, wherein the third conductive material includes at least one of titanium or copper.

[0108] Example 12 includes a package substrate comprising a glass core having a through-hole extending from a first surface of the glass core to a second surface of the glass core, a dielectric liner covering an inner surface of the through-hole, a conductive liner covering the dielectric liner, and a conductive fill material within the through-hole, the conductive liner including a different composition from the conductive fill material, the conductive liner separating the dielectric liner from the conductive fill material.

[0109] Example 13 includes the package substrate of example 12, wherein the conductive liner includes ruthenium.

[0110] Example 14 includes the package substrate of any one of examples 12 or 13, wherein a length of the through-hole is at least 10 times a width of the through-hole, the conductive liner conformally coating the dielectric liner with a substantially consistent thickness along the length of the through-hole.

[0111] Example 15 includes the package substrate of any one of examples 12-14, wherein the conductive liner includes less than 0.1wt % of titanium.

[0112] Example 16 includes the package substrate of any one of examples 12-15, wherein the dielectric liner includes an inorganic filler.

[0113] Example 17 includes the package substrate of any one of examples 12-16, wherein the dielectric liner includes at least one of an epoxy, a polyimide, or a parylene.

[0114] Example 18 includes the package substrate of any one of examples 12-17, wherein the dielectric liner includes silicon and at least one of oxygen, nitrogen, hydrogen, or carbon.

[0115] Example 19 includes a system comprising a semiconductor die, a package substrate including a glass core, the glass core having an opening with a height to width aspect ratio of at least 9, a first conductive material within the opening, and a second conductive material between the first conductive material and sidewalls of the opening, the second conductive material having a thickness that differs by less than 5% between any two points on the sidewalls of the opening, the second conductive material including ruthenium.

[0116] Example 20 includes the system of example 19, further including at least one of a keyboard or a display.

[0117] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

What is claimed is:

1. A package substrate comprising:

a core having a via extending therethrough;

a first conductive material in the via;

a dielectric material at least partially between a wall of the via and the first conductive material; and

a second conductive material in the via, the second conductive material closer to a central region of the via than the first conductive material is to the central region of the via, the first conductive material different from the second conductive material, the first conductive material at least partially between the dielectric material and the second conductive material.

2. The package substrate of claim **1**, wherein the core is a glass core.

3. The package substrate of claim **1**, wherein the first conductive material includes ruthenium, and the second conductive material includes copper.

4. The package substrate of claim **3**, wherein the first conductive material includes both ruthenium and copper.

5. The package substrate of claim **1**, wherein the via has an aspect ratio of at least 6.

6. The package substrate of claim **5**, wherein the first conductive material has a first thickness at a first end of the via, a second thickness at a second end of the via, and a third thickness at a midpoint of the via, the first thickness approximately equal to the second thickness and approximately equal to the third thickness.

7. The package substrate of claim **1**, wherein the dielectric material has a thickness between approximately 0.5 micrometers and 9 micrometers.

8. The package substrate of claim **1**, wherein the first conductive material has a thickness less than 70 nanometers.

9. The package substrate of claim **1**, further including a third conductive material at least partially between the first conductive material and the second conductive material, the third conductive material different from the first conductive material and different from the second conductive material.

10. The package substrate of claim **9**, wherein the third conductive material has a thickness between approximately 5 nanometers and 100 nanometers.

11. The package substrate of claim **9**, wherein the third conductive material includes at least one of titanium or copper.

12. A package substrate comprising:

a glass core having a through-hole extending from a first surface of the glass core to a second surface of the glass core;

a dielectric liner covering an inner surface of the through-hole;

a conductive liner covering the dielectric liner; and

a conductive fill material within the through-hole, the conductive liner including a different composition from the conductive fill material, the conductive liner separating the dielectric liner from the conductive fill material.

13. The package substrate of claim **12**, wherein the conductive liner includes ruthenium.

14. The package substrate of claim **12**, wherein a length of the through-hole is at least 10 times a width of the through-hole, the conductive liner conformally coating the dielectric liner with a substantially consistent thickness along the length of the through-hole.

15. The package substrate of claim **12**, wherein the conductive liner includes less than 0.1 wt % of titanium.

16. The package substrate of claim **12**, wherein the dielectric liner includes an inorganic filler.

17. The package substrate of claim **12**, wherein the dielectric liner includes at least one of an epoxy, a polyimide, or a parylene.

18. The package substrate of claim **12**, wherein the dielectric liner includes silicon and at least one of oxygen, nitrogen, hydrogen, or carbon.

19. A system comprising:

a semiconductor die;

a package substrate including a glass core, the glass core having an opening with a height to width aspect ratio of at least 9;

a first conductive material within the opening; and

a second conductive material between the first conductive material and sidewalls of the opening, the second conductive material having a thickness that differs by less than 5% between any two points on the sidewalls of the opening, the second conductive material including ruthenium.

20. The system of claim **19**, further including at least one of a keyboard or a display.

* * * * *